INF1008: Data Structures and Algorithms
Assignment 2
Evaluation of Data Structures or Algorithms using AI Chatbot

Objective:

The objective of this assignment is to evaluate a chosen data structure or algorithm using AI Chatbot. You will interact with Chatbot(s), ask it/them questions about your chosen topic, and assess its/their responses based on certain criteria, and implement an application that demonstrates the chosen data structure or algorithm.

Instructions:

1. **Choose a Data Structure or Algorithm**: Select a data structure or algorithm that you want to evaluate. It could be anything from a simple array or linked list to more complex structures like trees or graphs, or algorithms like sorting or searching algorithms.
2. **Interact with AI Chatbot(s)**: Use AI Chatbot(s) (e.g. OpenAI ChatGPT, Microsoft Copilot, Google Gemini, DeepSeek, etc.) to ask questions about your chosen data structure or algorithm. You can ask about its properties, use cases, advantages, disadvantages, time complexity, space complexity, etc.
3. **Evaluate the Chatbot's Responses**: Assess the chatbot's responses based on the following criteria:
   - **Accuracy**: Is the information provided by the Chatbot correct?
   - **Completeness**: Does the Chatbot provide a complete answer, or does it miss out on important details?
   - **Clarity**: Is the Chatbot's explanation clear and easy to understand?
   - **Relevance**: Does the Chatbot stay on topic and answer the question asked?
   - **Any other relevant evaluation criteria**
4. **Implement an Application**: Specify and implement a real-world application for your chosen data structure or algorithm in Python or a programming language approved by your tutor. Specify the functional components of the application and explain why your chosen data structure or algorithm is suitable for this application. Examples of a possible application can be a demo or illustration of the working of a data structure or algorithm (e.g. a maze search application that demonstrates a breadth-first search algorithm). You are encouraged to use AI Chatbot or other AI-based code generation tools such as CoPilot, Cursor or Windsurf to generate codes for the application. Explain your prompts clearly and what are the modifications added to the code (if any).

Submission for Assessment

1. Final report:
    - A report template in IEEE format is provided.
2. Package software:
    - A compressed (zipped format) file containing all the source files, software libraries (including open-source libraries), data files and instructions for the development and execution of the application. Please name the file: INF1008_ClassName_TeamXY.zip (e.g. INF1008_P1_Team01.zip).
3. Video submission:
    - Submit a video presentation of your project of a maximum of 10 minutes regardless of group size.
    - Submit a set of presentation slides used for the video presentation.
    - Upload the video to YouTube and submit the link to the YouTube video as a text file.
4. Complete online TeamMates peer evaluation (TeamMates link will be emailed to students by week 11).

Due Date: **End of Day 22nd March 2026**. Only one submission is required per group to xSiTe.

Report Format

1. Include the following sections in your report:
    - **Introduction**: Briefly introduce your chosen data structure or algorithm and the AI Chatbot(s) you used.
    - **Interactions**: Document the questions you asked (prompts) and the responses you received from the Chatbot(s).
    - **Evaluation**: Discuss your evaluation of the Chatbot’s responses based on the criteria mentioned above. Provide specific examples to support your evaluation.
    - **Application**: Discuss your implemented application for your chosen data structure or algorithm.
    - **Conclusion**: Summarize your findings and discuss any potential improvements that could be made to the Chatbot’s responses.

Presentation Format

1. Submit a video presentation of maximum 10 minutes regardless of group size. Your presentation should be engaging, well-structured, and demonstrate a deep understanding of the topic. Discuss your insights on the usefulness of the responses from AI Chatbots, and provide a demo on your application.

Rubrics:

- **Interactions with the Chatbot (15%)**: The quality and variety of questions asked to the Chatbot(s).
- **Evaluation of the Chatbot's Responses (20%)**: The depth and insightfulness of the evaluation.
- **Implement Application (30%)**: The specification, choice and implementation of the proposed application for the chosen data structure or algorithm.
- **Report Writing (20%)**: The clarity, organization, and presentation of the report.
- **Presentation (15%)**: The engagement, structure, and understanding demonstrated in the presentation.

Criteria	Excellent	Good	Satisfactory	Needs Improvement	Points
Interactions with the Chatbot	Asked a wide variety of insightful questions that thoroughly explored the topic.	Asked a good variety of questions that mostly explored the topic.	Asked a limited variety of questions that somewhat explored the topic.	Asked few or irrelevant questions that did not explore the topic.	15
Evaluation of the Chatbot's Responses	Provided a deep and insightful evaluation with specific examples.	Provided a good evaluation with some examples.	Provided a basic evaluation with few examples.	Provided a superficial evaluation with no or irrelevant examples.	20
Implement an Application	Implemented an application that is highly relevant, creative, complex, functional, and efficient for the chosen data structure or algorithm.	Implemented an application that is moderately relevant, creative, complex, functional, and efficient for the chosen data structure or algorithm.	Implemented an application that is somewhat relevant, creative, complex, functional, and efficient for the chosen data structure or algorithm.	Implemented an application that is irrelevant, uncreative, overly simple, nonfunctional, or inefficient for the chosen data structure or algorithm.	30
Report Writing	Report is clear, well organized, and professionally presented.	Report is mostly clear and organized, with minor issues in presentation.	Report is somewhat clear and organized, with some issues in presentation	Report is unclear or disorganized, with major issues in presentation.	20
Video Presentation	Presentation is engaging, well structured, and demonstrates a deep understanding of the topic.	Presentation is mostly engaging and structured, with a good understanding of the topic.	Presentation is somewhat engaging and structured, with a basic understanding of the topic.	Presentation is unengaging, poorly structured, or demonstrates a lack of understanding of the topic.	15

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